Length of a Line

Exam code: 0607

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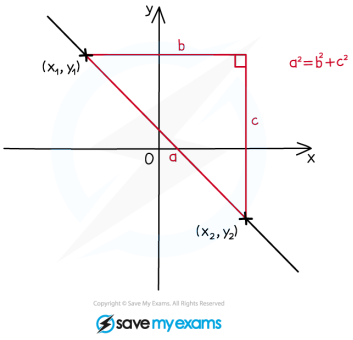
Length of a Line

How do I calculate the length of a line?

- The **distance between two points** with coordinates (x_1, y_1) and (x_2, y_2) can be found using the formula

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- This formula uses **Pythagoras' theorem** $a^2 + b^2 = c^2$
  - It is applied to the difference in the x-coordinates and the difference in the y-coordinates



Help me understand

Examiner Tips and Tricks

- Be extra careful when negative coordinates are involved
  - It can help to put negative numbers in brackets to make your working clearer
  - E.g. $(-6) - (-8) = 2$

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Worked Example

Point *A* has coordinates (3, -4) and point *B* has coordinates (-5, 2).

$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Substitute in the two given coordinates

$$d = \sqrt{(3-(-5))^2 + ((-4)-2)^2}$$

Be careful with the negative numbers
$3-(-5) = 8$ and $(-4)-2 = -6$

Simplify

$$d = \sqrt{(8)^2 + (-6)^2} = \sqrt{64+36} = \sqrt{100} = 10$$

10 units

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